

The empirical recurrence for OEIS A297430: recovering a conjecture that is not written down

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Abstract

OEIS A297430 records “Empirical recurrence of order 81”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 240$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 81, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 81 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297430 is “Number of $n \times 7$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 2 neighboring 1s.”. It has offset 1 and begins

1, 73, 301, 1057, 12005, 76692, 431723, 3338387, ...

The entry states:

Empirical recurrence of order 81 (see link above)

As of the “Last modified” line on the live entry (Dec 30 2017 07:00:01, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 16384$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 240$ of the 16384, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 240$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 480$ exact terms of the model returns order 81 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{81} c_i a(n-i),$$

c_1	3	c_{22}	43144295263	c_{43}	8882562049342	c_{64}	76661949942
c_2	19	c_{23}	−66967956411	c_{44}	−14052845894154	c_{65}	−61257326816
c_3	282	c_{24}	276728167312	c_{45}	−1793941737350	c_{66}	20832373511
c_4	−277	c_{25}	−201489300794	c_{46}	−3330509545750	c_{67}	−56645546749
c_5	−4323	c_{26}	558207559157	c_{47}	−7451446072213	c_{68}	545411339
c_6	−31610	c_{27}	−1298965850449	c_{48}	13499626620276	c_{69}	−1185130210
c_7	−21543	c_{28}	437376402109	c_{49}	−9734168689457	c_{70}	55168434
c_8	230400	c_{29}	−2149719598989	c_{50}	12798451789462	c_{71}	2262528498
c_9	1423701	c_{30}	4469096800795	c_{51}	−16026214065350	c_{72}	−275191754
c_{10}	1951784	c_{31}	257653062251	c_{52}	15173836748924	c_{73}	696186560
c_{11}	−3987095	c_{32}	3209551734948	c_{53}	−3788126754512	c_{74}	255822436
c_{12}	−25561797	c_{33}	−10451003786687	c_{54}	8793190522710	c_{75}	−37309676
c_{13}	−54862314	c_{34}	−3546792683370	c_{55}	−6105250394199	c_{76}	−44024820
c_{14}	2441494	c_{35}	3650738536618	c_{56}	−3270548974652	c_{77}	−12649444
c_{15}	108757691	c_{36}	13878810408331	c_{57}	−2080315842804	c_{78}	−2020996
c_{16}	738842718	c_{37}	8301783035749	c_{58}	−977959781193	c_{79}	−72608
c_{17}	277847769	c_{38}	−21193676288917	c_{59}	2384391102018	c_{80}	23232
c_{18}	2407452152	c_{39}	−6664679889985	c_{60}	19667208640	c_{81}	2304
c_{19}	−6356501597	c_{40}	−10175196115662	c_{61}	809191485586		
c_{20}	1908337349	c_{41}	29884000826178	c_{62}	−304846994577		
c_{21}	−38769393584	c_{42}	−2582772904066	c_{63}	14059432886		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 81, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $81 + 4$ terms removed returns order 81 as well, so $\deg q_{\min} = 81 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $81 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 81 that this sequence satisfies — and that family turns out to have exactly one member.

References

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